SMART CLASSROOM ANALYTICS

Research Paper Comparison Table

S.No	Paper Title	Author(s)	Year	Journal / Conference	Method / Technology Used	Advantages	Limitations
1	Learning Analytics for Smart Classrooms	**Cèlia Llurba, Ramon Palau, Jordi Mogas**	2025	**International Journal of Smart Education and Urban Society (IJSEUS)** – IGI Global (Book Chapter: *Learning Analytics for Smart Classrooms*)	Learning Analytics (LA), Artificial Intelligence, Big Data Analytics, Sensor Networks, Smart Classroom Technologies	• Supports personalized learning • Identifies at-risk students • Enhances teaching effectiveness • Enables data-driven decision-making • Improves student performance	• Privacy concerns • Limited adoption in schools • Requires high-quality educational data • High implementation cost
2	A Realtime Classroom Assessment System for Analysis of Students' Evaluation of Teaching Through a Deep Learning and Emotional Contagion Mechanism	**Kuan-Cheng Lin, Ya-Hsuan Lin, Mu-Yen Chen**	2025	**International Journal of Interactive Multimedia and Artificial Intelligence (IJIMAI)**	YOLO-based Deep Learning, Facial Emotion Recognition, Long Short-Term Memory (LSTM), Emotional Contagion Mechanism	• Real-time teaching evaluation • High emotion recognition accuracy • Automatic classroom assessment • Improves teaching quality • Enhances student engagement	• Privacy concerns • Requires high-quality cameras • Computationally intensive • Performance affected by lighting conditions
3	Classroom Student Behavior Recognition Using an Intelligent Sensing Framework	**Xian Min Zhao, Farrah Dina Yusop, Hai Chuan Liu, Yeni Nur Prilanita, Yi Xuan Chang**	2025	**IEEE Access**	Multimodal Framework Transformer (MFT), Computer Vision, Audio Processing, Transformer Networks	• Accurate student behavior recognition • Real-time monitoring • Multimodal learning • Personalized interventions • Improved classroom engagement	• Requires large datasets • High computational complexity • Sensor dependency • Privacy and security issues
4	Multimodal Deep Learning for Classroom Interaction Analysis Using Vision and Speech	**Xuan Wu**	2026	**Proceedings of the 2026 International Conference on Big Data and Informatization Education (ICBDIE 2026)**, ACM	CNN (ResNet-50), Bi-LSTM, Attention Mechanism, Vision and Speech Fusion, Deep Learning	• High interaction recognition accuracy • Combines visual and audio information • Better classroom analytics • Automated interaction assessment • Improves educational evaluation	• Computationally expensive • Requires synchronized multimodal data • Sensitive to background noise • Large training dataset required

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5	A Comparative Study of Creativity Among Senior Secondary Students in Smart and Traditional Classrooms in Ghaziabad	**Alka Chaudhary, Dr. Neeraj Kumar**	2025	**International Journal for Multidisciplinary Research (IJFMR)**	Smart Classroom Technologies, Interactive Learning Tools, Statistical Analysis (t-test)	• Enhances creativity • Encourages active learning • Better classroom participation • Improves critical thinking • Interactive learning environment	• Limited sample size • Region-specific study • Focuses only on creativity • Requires smart classroom infrastructure
6	Effects of Smart Classrooms on Learning and Teaching Effectiveness: The Students' Point of View	**Richard Marcellus, Omar Ghrayeb**	2025	**Proceedings of the ASEE Annual Conference & Exposition (American Society for Engineering Education)**	Multimedia Learning, Smart Classroom Technology, Student Survey Analysis	• Better visualization of concepts • Improves student engagement • Effective multimedia teaching • Easy presentation of learning materials • Enhances classroom experience	• Depends on instructor's teaching style • Limited survey participants • Technology dependence • Not suitable for every learning style
7	Intelligent Smart Classroom Learning Analytics Framework (DOI: s10758-025-09897-9)	**R. Sajja et al.**	2025	**Technology, Knowledge and Learning (Springer)**	Learning Analytics, Artificial Intelligence, Educational Data Mining, Smart Classroom Framework	• Efficient resource allocation • Improves teaching outcomes • Supports personalized education • Better educational decision-making • Data-driven learning	• Scalability challenges • Data privacy concerns • Infrastructure requirements • Complex implementation
8	Effects of Smart Classrooms on Learning and Teaching Effectiveness: The Students' Point of View	**Richard Marcellus, Omar Ghrayeb**	2025	**Proceedings of the ASEE Annual Conference & Exposition (American Society for Engineering Education)**	Smart Classroom Technology, Multimedia Instruction, Survey-Based Educational Analysis	• Improves classroom presentation • Better student engagement • Effective multimedia integration • Supports interactive learning • Enhances information delivery	• Limited respondent size • Technology dependence • Instructor adaptation required • Different learning preferences affect outcomes
9	Classroom Student Behavior Recognition Using an Intelligent Sensing Framework	**Xian Min Zhao, Farrah Dina Yusop, Hai Chuan Liu, Yeni Nur Prilanita, Yi Xuan Chang**	2025	**IEEE Access**	Multimodal Framework Transformer, Video & Audio Analysis, Computer Vision, Deep Learning	• Accurate behavior recognition • Real-time monitoring • Better classroom management • Personalized feedback • Supports intelligent education	• High computational requirements • Privacy concerns • Requires multimodal sensors • Dataset dependency

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10	Multimodal Deep Learning for Classroom Interaction Analysis Using Vision and Speech	**Xuan Wu**	2026	**Proceedings of the 2026 International Conference on Big Data and Informatization Education (ICBDIE 2026)**, ACM	CNN (ResNet-50), Bi-LSTM, Attention Mechanism, Speech Recognition, Deep Learning	• High recognition accuracy • Vision and speech fusion • Automated interaction analysis • Better educational assessment • Improved engagement detection	• High computational cost • Noise sensitivity • Requires synchronized data • Large-scale dataset needed
11	A Comparative Study of Creativity Among Senior Secondary Students in Smart and Traditional Classrooms in Ghaziabad	**Alka Chaudhary, Dr. Neeraj Kumar**	2025	**International Journal for Multidisciplinary Research (IJFMR)**	Smart Classroom Technologies, Interactive Learning, Statistical Analysis (t-test), Educational Research	• Enhances creativity • Encourages innovative thinking • Better student participation • Improves learning outcomes • Promotes interactive education	• Limited geographical coverage • Small sample size • Infrastructure dependency • Focuses only on creativity

RESEARCH IDENTIFICATION AND JUSTIFICATION:-

The eleven sources under review collectively span the empirical, technical, affective, perceptual, and analytical dimensions of smart cl assroom scholarship, thereby furnishing a reasonably comprehensive evidentiary foundation. Chaudhary and Kumar (2025) contribute an empirical, t-test-based comparison of creativity indices—fluency, flexibility, and originality—between smart and traditional classrooms among CBSE Class 11 and 12 students in Ghaziabad, offering direct quantitative support for the proposition that technologically enhanced environments foster divergent thinking, while also disaggregating outcomes by gender and academic stream. Wu (2026) advances the technical dimension of the field through a multimodal deep learning architecture that fuses ResNet -50 visual features with BiLSTM-processed speech via a cross-modal attention mechanism, demonstrating the feasibility of automated, sensor-driven engagement and interaction-pattern recognition in real classroom settings. Lin, Lin, and Chen (2025) extend this technical trajectory into the affective domain, employing YOLO-based facial emotion recognition coupled with an LSTM model to predict teaching-satisfaction scores through an emotional-contagion mechanism, thereby offering a methodologically rigorous alternative to conventional self-reported teaching evaluations. Huang (2025) similarly contributes to the computer-vision strand of the literature by proposing an enhanced YOLO11-DP model augmented with a mixed local channel attention mechanism for real-time student behaviour detection, supplying benchmarked performance metrics that substantiate claims regarding the maturity of deep -learning-based classroom monitoring. Marcellus and Ghrayeb (2002), by contrast, provide an earlier and deliberately student-centred qualitative account, drawing on survey responses and reflective essays to nuance the assumption that technology-mediated instruction is uniformly superior to blackboard pedagogy, and are therefore valuable as a counterbalancing historical reference. Uskov et al. (2020) contribute methodological depth on the data-analytics side, systematising five distinct forms of student academic performance data representation and benchmarking multiple machine-learning models for predictive accuracy, which is pertinent wherever a study engages with learning-analytics infrastructure. Finally, Llurba, Palau, and Mogas (2023) function as a review-of-reviews, synthesising thirty-five studies on learning analytics in smart classrooms through a PRISMA-based systematic methodology, and are useful for situating a given study within the broader field while also surfacing acknowledged gaps such as the underrepresentation of primary- and secondary-school contexts and unresolved data-privacy concerns.

SDG MAPPING:-

Across all seven sources, the most consistent point of convergence is SDG 4 (Quality Education), reflecting the shared preoccupation with improving pedagogical effectiveness, learner engagement, and instructional quality through technological means. Beyond this common thread, several sources extend into adjacent goals: Chaudhary and Kumar (2025) additionally engage SDG 5 (Gender Equality) through their g ender-disaggregated analysis of creativity outcomes; Wu (2026) and Huang (2025) both align with SDG 9 (Industry, Innovation and Inf rastructure) by virtue of their contributions to AI-driven educational infrastructure; Lin, Lin, and Chen (2025) intersect with SDG 3 (Good Health and Well-Being) insofar as their emotional-contagion framework links teaching quality to the psychological well-being of both teachers and students; Marcellus and Ghrayeb (2002) remain centred on SDG 4 alone, given their focus on equitable and effective learning experience rather than infrastructural innovation; Uskov e t al. (2020) again engage SDG 9 through their predictive learning-analytics architecture; and Llurba, Palau, and Mogas (2023) additionally touch upon SDG 10 (Reduced Inequalities) and SDG 16 (Peace, Justice and Strong I nstitutions) by virtue of their sustained attention to equity of access and the ethical and dataprivacy dimensions of learning-analytics adoption in institutional settings.

REFERENCES:-

- Llurba, C., Palau, R., & Mogas, J. (2023). *Learning analytics for smart classrooms*. In *International Journal of Smart Education and Urban Society*. IGI Global. https://doi.org/10.4018/978-1-7998-9220-5.ch103
- Lin, K.-C., Lin, Y.-H., & Chen, M.-Y. (2025). *A realtime classroom assessment system for analysis of students' evaluation of teaching through a deep learning and emotional contagion mechanism. International Journal of Interactive Multimedia and Artificial Intelligence, 9*(5), 51–59. https://doi.org/10.9781/IJIMAI.2025.03.002
- Zhao, X. M., Yusop, F. D., Liu, H. C., Prilanita, Y. N., & Chang, Y. X. (2025). *Classroom student behavior recognition using an intelligent sensing framework. IEEE Access, 13*, 49767–49776. https://doi.org/10.1109/ACCESS.2025.3550921
- Wu, X. (2026). *Multimodal deep learning for classroom interaction analysis using vision and speech*. In *Proceedings of the 2026 International Conference on Big Data and Informatization Education (ICBDIE 2026)* (pp. 1–6). ACM. https://doi.org/10.1145/3806980.3807070
- Chaudhary, A., & Kumar, N. (2025). *A comparative study of creativity among senior secondary students in smart and traditional classrooms in Ghaziabad. International Journal for Multidisciplinary Research (IJFMR), 7*(3), 1–10.
- Marcellus, R., & Ghrayeb, O. (2002). *Effects of smart classrooms on learning and teaching effectiveness: The students' point of view*. In *Proceedings of the American Society for Engineering Education (ASEE) Annual Conference and Exposition*.
- Sajja, R., Sermet, Y., Cwiertny, D., & Demir, I. (2025). *Integrating AI and learning analytics for data-driven pedagogical decisions and personalized interventions in education. Technology, Knowledge and Learning*. https://doi.org/10.1007/s10758-025-09897-9

- Marcellus, R., & Ghrayeb, O. (2002). *Effects of smart classrooms on learning and teaching effectiveness: The students' point of view*. In *Proceedings of the American Society for Engineering Education (ASEE) Annual Conference and Exposition*.
- Zhao, X. M., Yusop, F. D., Liu, H. C., Prilanita, Y. N., & Chang, Y. X. (2025). *Classroom student behavior recognition using an intelligent sensing framework. IEEE Access, 13*, 49767–49776. https://doi.org/10.1109/ACCESS.2025.3550921
- Wu, X. (2026). *Multimodal deep learning for classroom interaction analysis using vision and speech*. In *Proceedings of the 2026 International Conference on Big Data and Informatization Education (ICBDIE 2026)* (pp. 1–6). ACM. https://doi.org/10.1145/3806980.3807070
- Chaudhary, A., & Kumar, N. (2025). *A comparative study of creativity among senior secondary students in smart and traditional classrooms in Ghaziabad. International Journal for Multidisciplinary Research (IJFMR), 7*(3), 1–10.